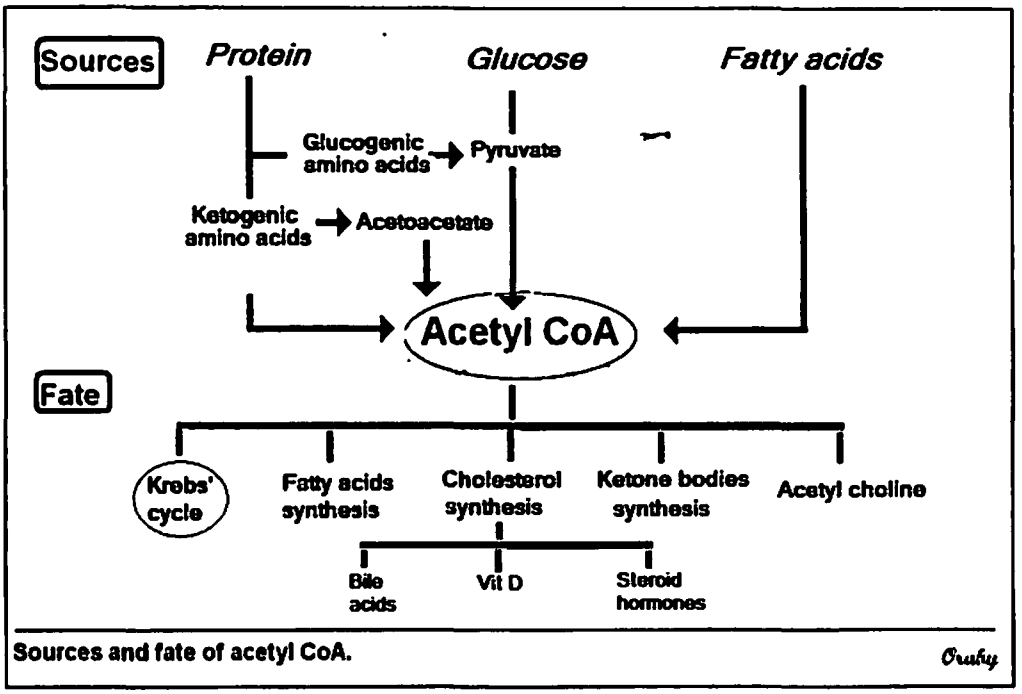


Sources and fate of acetyl CoA.

FATTY ACIDS SYNTHESIS

I. INTRODUCTION:

A considerable number of fatty acids are derived from diet. Living organisms have the capacity to synthesize fatty acids from acetyl CoA. Any substance that gives acetyl CoA (e.g. glucose) is called lipogenic.

There are 2 mechanisms for fatty acids synthesis: cytoplasmic and microsomal.

II. CYTOPLASMIC SYSTEM FOR FATTY ACID SYNTHESIS: Also called extramitochondrial system or de novo synthesis of fatty acids. The main product of this pathway is **palmitate (16 C)**.

A. **Site:**

1. **Intracellular location:** Cytosol.
2. **Organ location:** Many tissues including liver, adipose tissue, mammary gland, lung and kidney.

B. **Requirements:** This pathway needs the following substrates; acetyl CoA, $NADPH+H^+$ and group of enzymes called collectively fatty acid synthase complex.

1. **Acetyl CoA:**

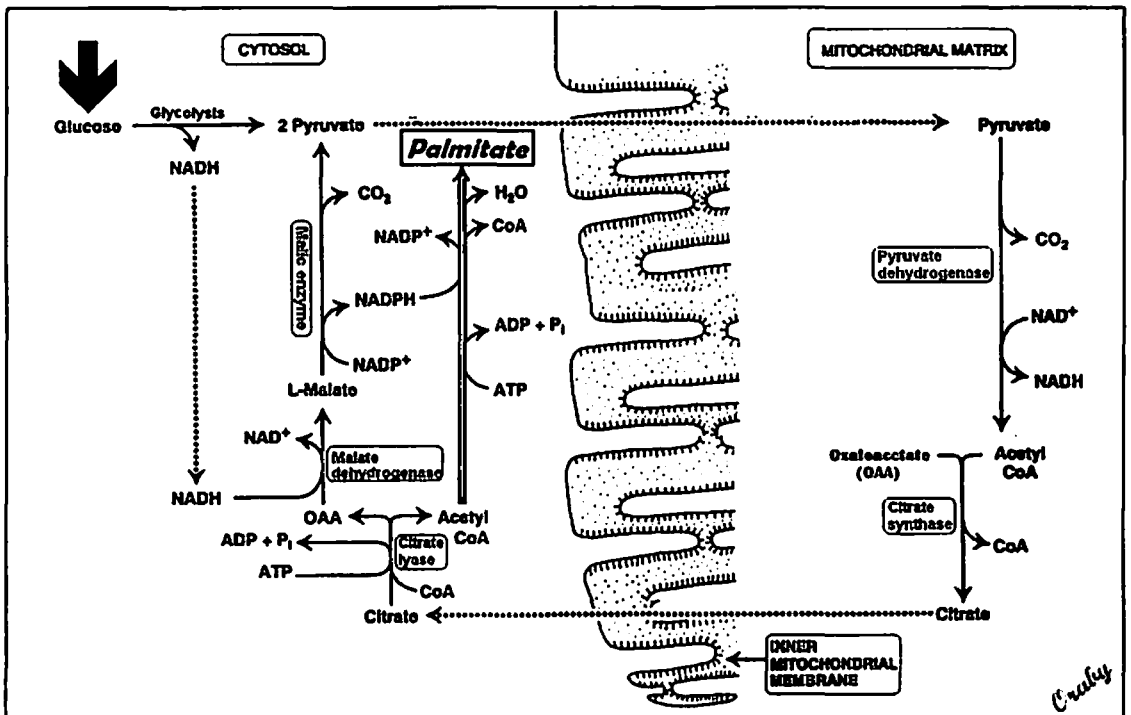
a) It is provided mainly by glucose through pyruvate (glycolysis).

b) Acetyl CoA is formed in mitochondria and fatty acid synthesis occurs in cytosol. The acetyl CoA cannot diffuse to cytosol because mitochondrial membrane is impermeable to it. Acetyl CoA condenses

with oxaloacetate – in the presence of citrate synthase- to form citrate. Then citrate diffuses out of mitochondria to cytosol where it is splitted again – by citrate lyase – into acetyl CoA and oxalaoacetate (see the following figure):

1) Acetyl CoA molecules are used for palmitate synthesis.
2) Oxaloacetate is converted to malate: Malate may be converted to pyruvate by **malic enzyme.** This reaction produces $NADPH+H^+$, which is needed for fatty acid synthesis.

Note: *Acetyl CoA used for fatty acid synthesis always derived from glucose and never from fatty acids. This is because insulin hormone secreted after meal* **stimulates** *both glucose oxidation (➔acetyl CoA) and lipogenesis (=Fatty acid synthesis) and* **inhibits** *lipolysis (➔Fatty acid oxidation➔Acetyl CoA).*



Interrelationship between glucose metabolism and palmitate synthesis. [Note: OAA= Oxaloacetate].

2. **$NADPH+H^+$:** It is provided by 3 sources:
   a) Pentose phosphate pathway.
   b) Action of cytoplasmic isocitrate dehydrogenase on isocitrate. It is similar to mitochondrial one but it uses $NADP^+$ as hydrogen carrier.

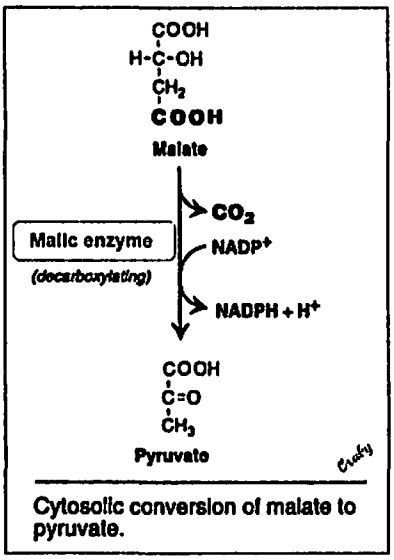
Isocitrate+$NADP^+$➔Oxalosuccinate + $NADPH+H^+$ ➔ α Ketoglutrate

   c) Action of malic enzyme on malate to produce pyruvate.
3. **Fatty acid synthase complex:**
   a) This enzyme is a **dimer** i.e. formed of 2 subunits.

b) Each unit, which is called **monomer**, contains 7 enzymes and a terminal protein called **acyl carrier protein (ACP).**

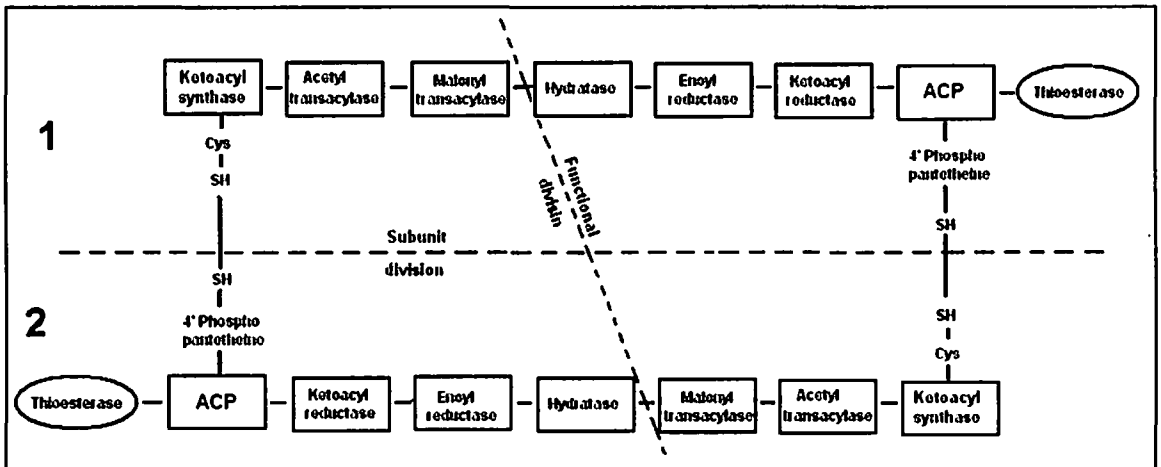
c) ACP is a protein contains the vitamin pantothenic acid in the form of phosphopantotheine. ACP is the part that carries the acyl group.

d) Each monomer contains 2 –SH groups, one provided by phosphopantotheine and attached to ACP. The other is provided by cysteine attached to the enzyme 3-ketoacyl synthase.



Cytosolic conversion of malate to pyruvate.

e) The 2 monomers are arranged head to tail, so the -SH group of ACP of one monomer is very close to the –SH group provided by 3-ketoacyl synthase of the other monomer.



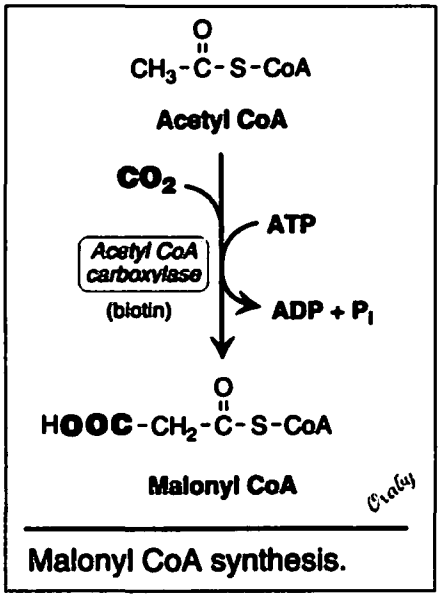
Fatty acid synthase complex. The active enzyme is a dimer of identical subunits.

C. Steps of cytoplasmic pathway:

1. **Carboxylation of acetyl CoA to form malonyl CoA:**

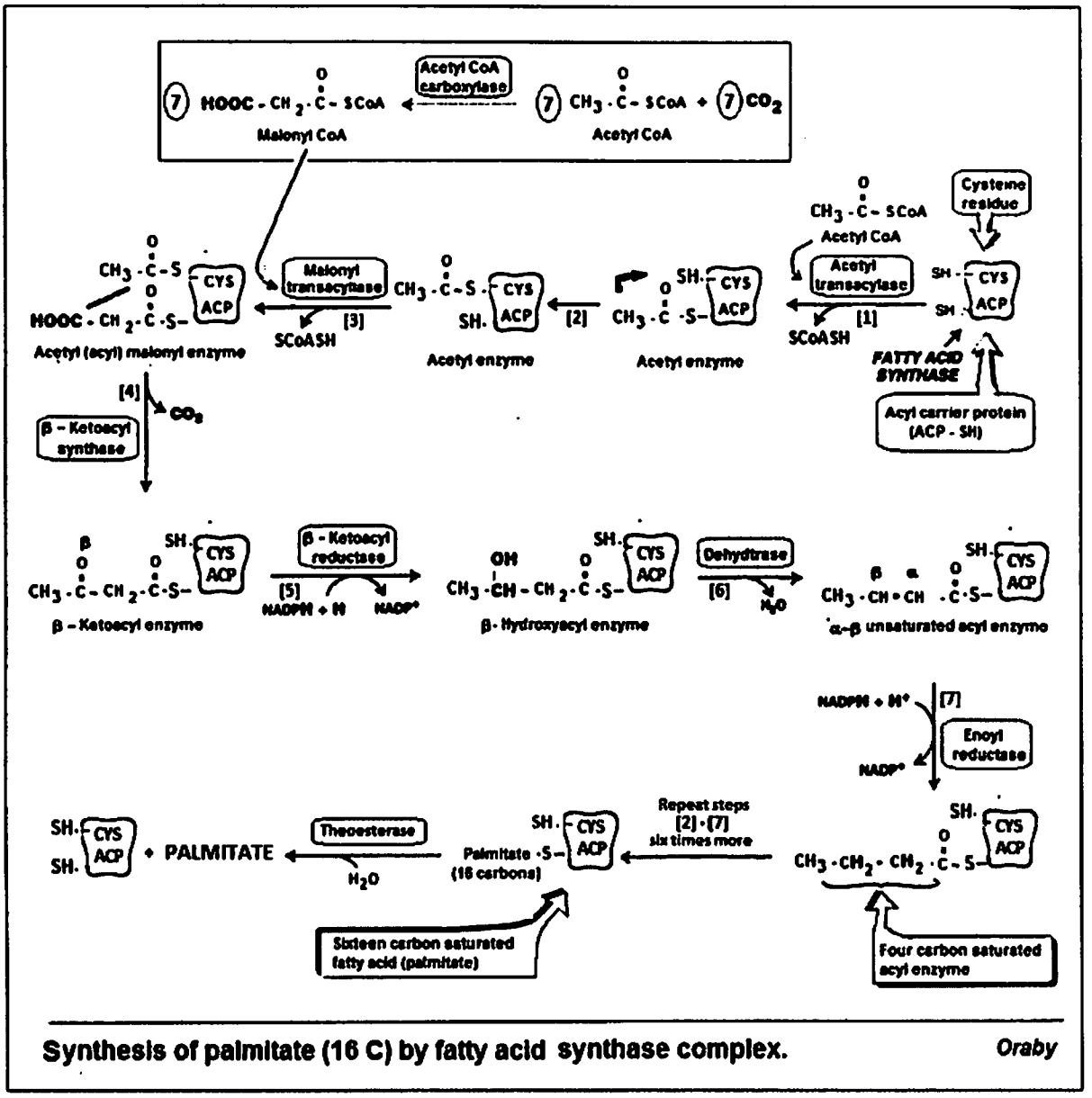
a) Malonyl CoA is synthesized from acetyl CoA by **acetyl CoA carboxylase** in the presence of biotin and ATP.

b) **It is the committed step in the pathway.**



Malonyl CoA synthesis.

2. **Synthesis of palmitate:**

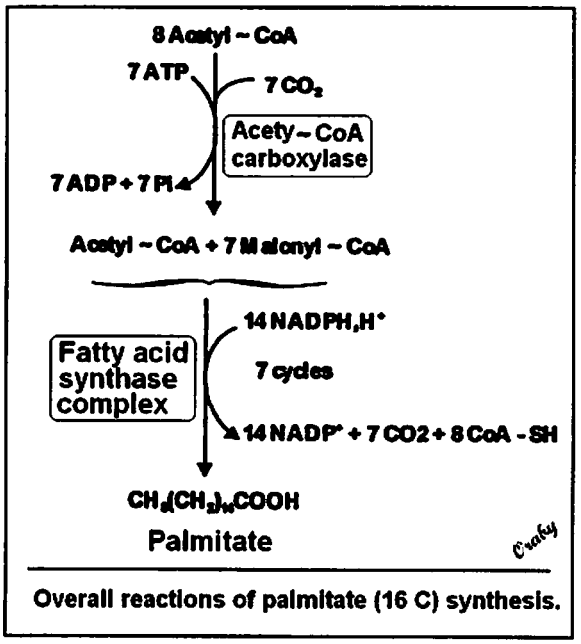


Synthesis of palmitate (16 C) by fatty acid synthase complex. *Oraby*

D. <u>Comments on synthesis of palmitate</u>:

1. A molecule of acetyl CoA combines with a pantotheine –SH group of the first monomer. This step is catalyzed by acetyl (acyl) transacylase enzyme. Then acetyl group is transferred to the cysteine –SH group of the other monomer of fatty acid synthase

2. A molecule of malonyl CoA then combines with a pantotheine –SH group of the first monomer. This step is catalyzed by malonyl transacylase enzyme. This leads to the formation of acetyl (acyl)-malonyl-enzyme.
3. Then acetyl group attacks the malonyl group to liberate CO_2 and to form 3 ketoacyl enzyme (acetoacetyl enzyme) that attached to pantotheine–SH group and to let the cysteine–SH group free (which was initially occupied by the acetyl CoA). *This decarboxylation step acts as a driving force for the whole reactions.*
4. The 3-ketoacyl group is reduced, dehydrated and then reduced again to form the corresponding acyl-enzyme (4 carbon saturated fatty acyl enzyme)
5. The 4 carbons saturated fatty acyl group is then transferred to cysteine –SH group of the other monomer of fatty acid synthase leaving the pantotheine –SH group of the first monomer for the second molecule of malonyl CoA.
6. The sequence of reactions is then repeated 6 more times to produce fatty acids of 6,8,10,12,14 and finally 16 carbons (palmitic acid).
7. Free palmitate is formed by the action of thioesterase enzyme of the multi-enzyme complex, which adds H_2O and hydrolyzes the palmitoyl enzyme.
8. **The overall reaction is as shown in the diagram:**



Overall reactions of palmitate (16 C) synthesis.

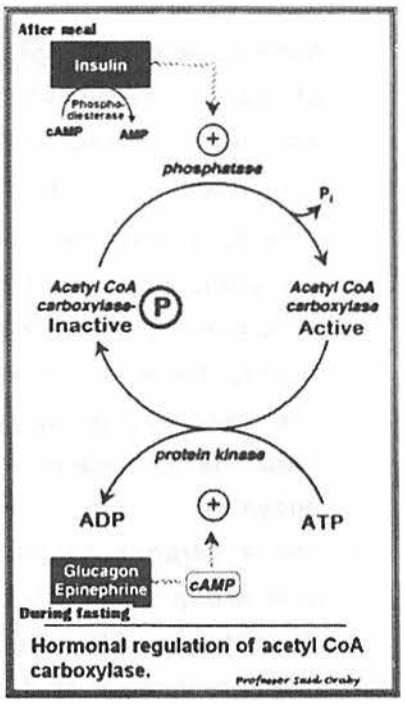
E. Fate of palmitate:

1. **Esterification:**
   Palmitate esterifieid with glycerol to from acylglycerols or with cholesterol to form cholesteryl ester.
2. **Chain elongation:**
   Palmitate may be elongated to form a longer fatty acid.
3. **Desaturation:** i.e. synthesis of unsaturated fatty acid: palmitate may undergo desaturation at C_9-C_{10} to form palmitoleic acid.
4. **Sphingosine formation:**
   It is formed by condensation of palmityl CoA and the amino acid serine.

F. **Regulation of fatty acid synthesis:**

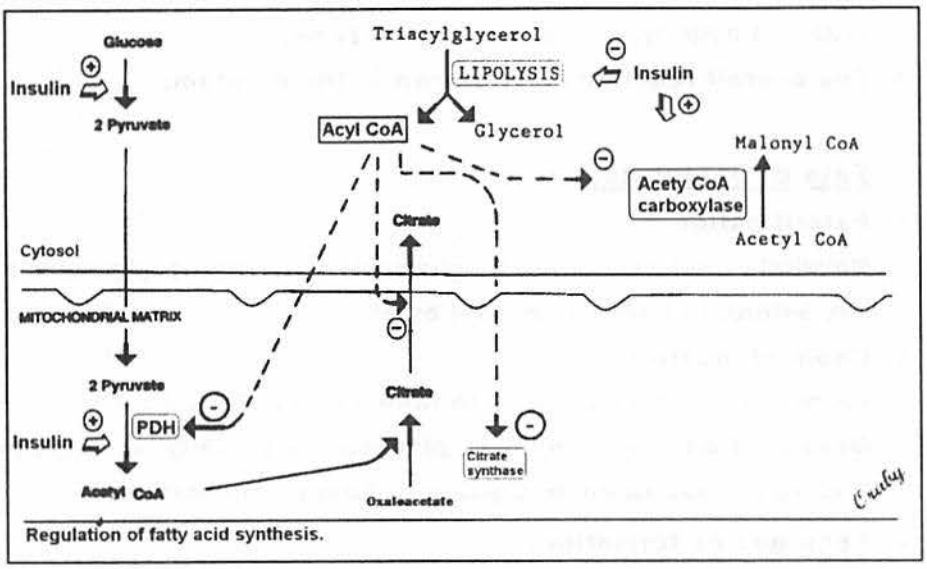
The key enzyme of cytoplasmic pathway is acetyl CoA carboxylase enzyme.

1. **Insulin:** Stimulates fatty acid synthesis through:
   a) Insulin activates both acetyl CoA carboxylase and pyruvate dehydrogenase [PDH] complex (through dephosphorylation of the enzymes).
   b) Insulin stimulates also the transport of glucose into cells e.g. adipose tissue. This increases the amount of pyruvate ⇒ Acetyl CoA ⇒ Fatty acid synthesis.
   c) Insulin inhibits lipolysis through inhibition of cAMP ⇒ No long chain fatty acids (acyl CoA) that inhibit lipogenesis.



Hormonal regulation of acetyl CoA carboxylase.

2. **Glucagon and epinephrine:** Inhibits fatty acid synthesis through cyclic AMP and it also stimulates lipolysis ➔ Long chain fatty acids ➔ inhibition of lipogenesis.
3. **Long chain acyl CoA**: Inhibits Fatty acid synthesis through:
   a) It inhibits allosterically acetyl CoA carboxylase.
   b) It inhibits transport of citrate from mitochondria to cytosol.
   c) It inhibits pyruvate dehydrogenase (PDH) that synthesizes pyruvate ➔ Acetyl CoA ➔ fatty acid.
4. **Citrate:** stimulates fatty acid synthesis through stimulation of acetyl CoA carboxylase.



Regulation of fatty acid synthesis.

	FA acid synthesis	FA acid oxidation
Organ location	Mainly liver	Mainly liver and muscles
Intracellular location	Cytosol	Mitochondria
Nutritional state	After meals	Fasting
Hormonal stimulation	Insulin	Glucagon
Carrier of acetyl/acyl group between cytosol and mitochondria	Citrate (mitochondria to cytosol).	Carnitin (cytosol to mitochondria)
Active carriers	➢ Acyl carrier protein ➢ CoASH	➢ CoASH
Coenzyme	$NADP^+$	NAD^+
Two carbon donor / product	Malonyl CoA	Acetyl CoA
Activator	Citrate	
Inhibitor	Fatty acyl CoA (inhibits acetyl CoA carboxylase)	Malonyl CoA (inhibits Carnitine acyl transferase)
Product of pathway	Palmitate	Acetyl CoA

Comparison between fatty acid synthesis (extramitochondrial pathway) and fatty acid oxidation

Professor Said Orabi

III. *MICROSOMAL PATHWAY FOR FATTY ACID SYNTHESIS:*

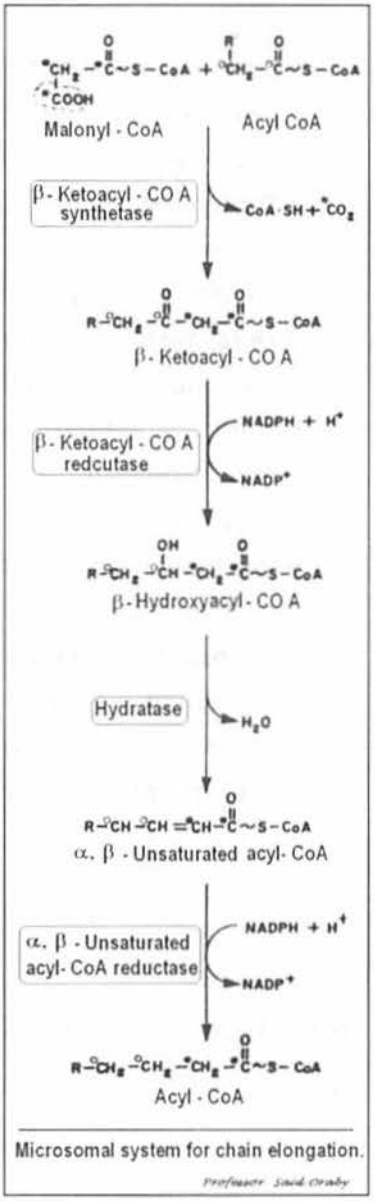
This is probably the main site for the elongation of existing long chain fatty acid molecules i.e. production of fatty acids longer than 16 carbon atoms.

A. <u>The elongated molecules (C10-C16) are derived from:</u>

1. Palmitate: by cytoplasmic pathway.
2. fatty acids of diet.

B. The microsomal pathway needs malonyl CoA as acetyl donor and $NADPH+H^+$ as coenzyme.

C. <u>Function</u>: This system becomes active during myelination of nerves in order to provide C_{22} and C_{24} fatty acids which are present in sphingolipids.



Microsomal system for chain elongation.

Professor Said Orabi

IV. *Synthesis Of Unsaturated Fatty Acids:*

A. <u>Nonessential unsaturated fatty acids:</u>

1. These are fatty acids which contain one double bond e.g. palmitoleic acid (16:1) and oleic acid (18:1).
2. **Synthesis of oleic acid (oleyl CoA):** It is synthesized – in the microsomes - from stearyl CoA (active stearic acid).

B. <u>Essential fatty acid:</u> These are unsaturated fatty acids which contain more than one double bond.

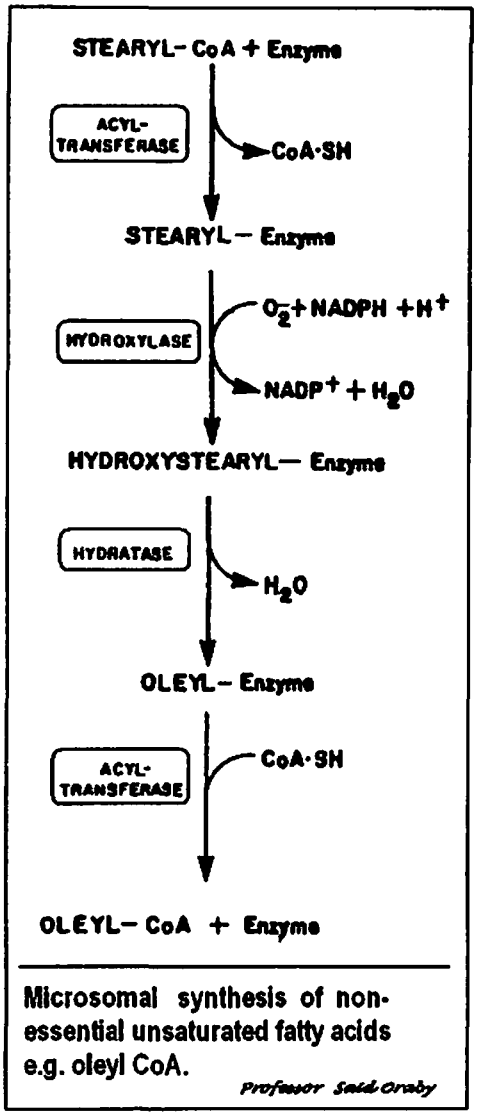
1. They are essential because they cannot be formed in the body and should be taken in the diet
2. **Examples:** linoleic acid (18:2) linolenic (18:3) and arachidonic acid (20:4).
3. **Sources:** Vegetable oils as corn oil and cotton seed oil.
4. **Functions:**

a- They are important for **normal growth.**

b- Synthesis of phospholipids: They enter in the structure of Phospholipids mainly in the 2nd position. Phospholipids have many functions as:

1) Enter in the structure of cell membranes.
2) They act as lipotropic factors i.e. prevent accumulation of fat in liver.
3) Dipalmityl lecithin acts as surfactant in lungs.
4) Cephalin is important for coagulation.

c- Prevention of atherosclerosis: Essential fatty acids combine with cholesterol forming esters which are rapidly metabolized by the liver. This prevents precipitation of free cholesterol along the endothelium of blood vessels ⇒ prevents atherosclerosis.



Microsomal synthesis of non-essential unsaturated fatty acids e.g. oleyl CoA.

Professor Said Oraby

d- Synthesis of eicosanoids: Arachidonic acid gives rise to a group of compounds called **eicosanoids**. They comprise the prostanoids, leukotrienes (LTs) and lipoxins (LXs). Prostanoids include prostaglandins (PGs), prostcyclins (PGIs) and thromboxane TXs).

EICOSANOIDS

I. **<u>Definition:</u>** These are cyclic compounds that derived from arachidonic acid (eicosatetraenoic) (20 C) after cyclization of its carbons chain to form a ring.